



Guidelines on Unattended/Autonomous Experiments in the Laboratory

Scope

Unattended or autonomous experiments are experiments where the apparatus for the experiment has been left to run while the people responsible for the operation of the experiment are not in the area to monitor events and no one has been assigned to cover the absence of the main experimenter. In some cases, there may be remote monitoring or logging in process, however this is still considered unattended or autonomous. In some labs there may be robotic equipment that is conducting experiments, either in a contained area like a stationary robot, or in some cases with free moving robotics that can independently work with equipment such as fume hoods in the open lab.

This document provides a framework for risk assessment and does not replace local risk assessment.

By the end of this document, you should be able to:

- Understand what responsibilities are for; the Chair, the Principal Investigator, students/staff and Environmental Health and Safety (EHS).
- Understand what safety considerations must be looked at when embarking on experiments that will be unattended. This will include criteria for inclusion/exclusion of experiments with respect to being run unattended.
- Understand what types of experiments may need to be considered for further mitigation of risk when being run unattended.
- Understand what notification needs to be posted on the exterior lab door.
- Understand what the minimum requirements for an unattended operations standard operating procedure (SOP) would be.
- Understand how to assess what other requirements may need to be fulfilled.

Definitions

- Unattended: Experiments running for extended periods while no lab member is assigned oversight of the experiment.
- Automated: Tasks executed by machines, where each cycle is started by a human.
- Autonomous: Workflows initiated and executed entirely by the system without human intervention.
- Release: An event, including but not limited to, a water leak, spill of hazardous liquid, release of vapour, fire or explosion, etc.



Responsibilities

Department Chair/equivalent or their Designate is responsible for:

- Informing the PI of any interruptions in service of which they are aware that may interrupt unattended experiments.
- Having a process as applicable for working with PI's to implement this Guideline.

Principal Investigators (PIs)/Designate are responsible for:

- Providing any required approval for unattended or autonomous experiments and equipment.
- Reviewing and assessing potential releases or incidents that could lead to harm or property damage.
- Advising the department of unattended operations that are occurring in their lab.
- Consulting with the department and seeking approval from the chair as applicable for high or medium risk activities (see sections 5 and 6) where the consequences of a failure cannot be kept within the containment zone.
- Ensuring that any activities are as safe as is reasonably achievable in an experimental environment.
- Ensuring that an emergency contact is available during the duration of the experiment.
- Ensuring that any needed signage is posted so that staff and students are properly informed of any risks presented by the activities.

Students/Staff conducting Unattended/Autonomous Activities are responsible for:

- Consulting with the PI on any planned unattended or autonomous experiments.
- Following any instructions from the PI on the planned experiments.
- Following all applicable SOPs, manufacturers recommendations and industry standards as applicable.
- Following these guidelines.

Environmental Health and Safety (EHS):

- Shall update this standard periodically as needed.
- Assist Departments/Institutes with local risk assessment when needed; Incorporate industry standards and best practices to update this guideline.

Considerations

Unattended Operations:

- Are discouraged except where necessary and with approval of the PI (Principal Investigator).
- Should be a well understood process.
- Should not be conducted unattended if there is a high chance of adverse outcome or a high hazard in the event of a release and/or incident
- Must be approved by the PI or designate who knows the work and the implications of a system failure best.
- May be routine and low risk. These operations can be covered by a lab SOP.



- May have considerations and precautions that are unique.
- Must be assessed for risk and mitigation – please use the Risk Assessment Tool as a guideline and note that some experiments may have issues not specifically covered in the tool.
- Must have secondary containment sufficient to hold the full projected volume of the experiment without overflowing. For instance, a tray to contain any liquids that may escape.
- Must have at least a sign posted in the lab to inform others in the lab that an autonomous or unattended experiment is in progress and the hazards presented by the operation. The sign must be near the site of operations and should be large enough that basic information can be read from a safe distance. See sample sign in appendix.
- Some operations may also require a sign on the door alerting after hours personnel such as caretakers, maintenance personnel and Campus Security that there are unattended operations going on inside and with brief instructions on what to do if ventilation fails or an alarm sounds. An after-hours phone number for a knowledgeable lab person must be available to Campus Security, although it does not need to be on the sign if the number has been supplied to Campus Security in advance. See sample sign in appendix.
- Have an emergency shutoff wired in for any autonomously moving equipment.
- Keep the area around the experiment clear of hazardous or combustible (fuel) materials – no waste containers, reagents, solvents, boxes etc.
- Consult the [Guideline on University Built Equipment](#) for any equipment that is designed or built by the experimenters.

At a minimum include the following checks as part of your SOP before beginning:

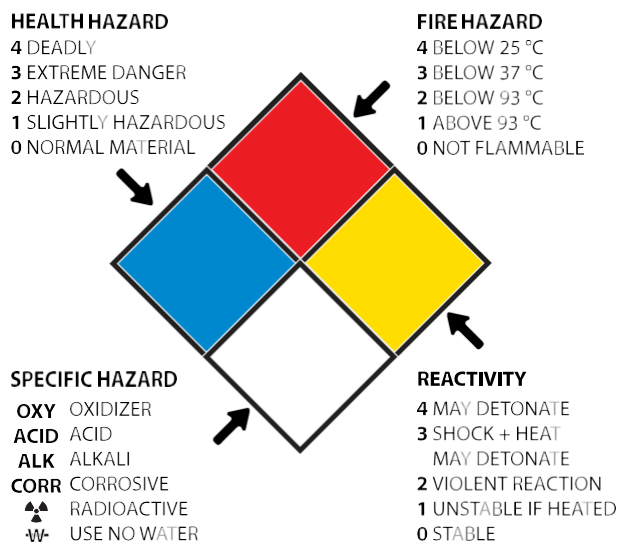
- Equipment is in good condition and operating within normal parameters – check logbook as needed.
- Any water connection.
- The volume remaining in any inert gas supply
- Available space for effluent in any collection or overflow containers
- Condition of hoses or pipe connections.
- Condition of electrical connections.
- Area is clear of any items that could burn or spread a fire.
- Containment is adequate for the volume of the experiment.
- The area is clean and uncluttered.
- Any combustible or self-igniting materials are localized and away from heat/spark sources.



Risk Assessment Tool for Chemicals

The consequence of any release of materials is influenced by their volatility (the tendency of the materials to become airborne), the flammability, the reactivity, and the toxicity. The NFPA fire diamonds found on SDSs can be used to give a quick assessment of the risk presented by any release.

For any given material that might be released, rate the severity of the release by taking the NFPA 704 (fire diamond) numbers for flammability, reactivity, and health from the diamond on the SDS or label and sum them.



(Image courtesy Marcos Rodríguez Bobadilla, via Wikimedia Commons)

If the sum of the numbers is 6 or more consider the material to be HIGH RISK.

If the sum of the numbers is 4-5 consider the material to be MEDIUM RISK.

If the sum of the numbers is 3 or lower consider the material to be LOW RISK.

Risk Assessment for Biologicals

For any given material that might be released, risk is based on the containment level assigned for the material. CL2 and above is High Risk, CL1 is low risk.



Loss of Services Assessment

Loss of Service ("What If")	Mitigation	Comments
Electricity		Consider use of red emergency power outlet
Cooling Water		Consider closed loop cooling system
Other loss of cooling		
Room ventilation shut-down due to loss of power	At a minimum there must be a sign on the door warning of potential airborne hazards in the room in the event of ventilation failure – see Appendix.	
Fume hood failure		
Loss of inert atmosphere due to gas running out or leak in system		
Stirrer bar failure		
Heat source failure		
Loss of vacuum		
Other;		
Other;		
Other;		

Signage

If the result of the assessment is that the experiment is High or Medium risk to people or to property in the event of a failure, a door sign must be posted warning potential first responders caretakers, or maintenance personnel that the activities in the room may present a risk.

Overall Assessment of Risk to People for Door Sign (circle one):

High

Medium

Low (no sign needed)

Overall Assessment of Risk to Property for Door Sign (circle one):

High

Medium

Low (no sign needed)

PI or designate signature_____

Name of signer_____



Appendix A – Signage

Door Sign

Door sign should clearly display information from applicable sections on this page that are relevant to operations in progress.

Warning Unattended Operation in Progress

Risk to Health
High Risk to Health on Ventilation Failure
Potential for significant release of highly toxic, highly flammable, or other highly hazardous materials into the room air if the fume hood fails. There is greater risk in the event of both fume hood and room ventilation failure.
Potential for Fire due to air or water reactive materials.
Potential for out-of-control reaction.
Medium Risk
There is a moderate risk to health if fume hood or laboratory ventilation fails. Moderately toxic materials may be released to the air or smaller amounts of vapours from flammable or combustible materials.

Risk to Property
Continuously Running Water (e.g., equipment cooling water).
Tank or reaction vessel of water or other non-hazardous material – e.g. non-pathogenic yeast.
Risk of leaks (loss of containment).
Potential for non-hazardous or low hazard odours being emitted.

Contact Information
Name:
After Hours Phone Number:
Name:
After Hours Phone Number:

Please note for contact information, a personal cell phone is not required as long as the number provided will work in an emergency. For example, a number that forwards could be used, or the number of a responsible person that can contact the experimenter. If Campus Security has the emergency number, the Campus Security number could be used here.



Sign Near Source of Hazard

Note actions to control processes are for trained lab staff only – first responders should contact lab staff. First responders are not responsible for “saving” experiments.

Description of Experiment in Progress:

List of Potential Problems with Experiment (e.g. overheating, overflowing, excess pressure, loss of power, loss of containment, leaks etc.):

1. _____
2. _____
3. _____
4. _____

Emergency Procedure to Shut Down Operation:

1. _____
2. _____
3. _____
4. _____

Experiment Owner:

Contact Information:

Please note for contact information, a personal cell phone is not required as long as the number provided will work in an emergency. For example, a number that forwards could be used, or the number of a responsible person that can contact the experimenter.